

Nihar Ranjan Samantaray

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SUMMARY

Software Engineer with 5.8+ years of experience in building scalable data pipelines and cloud-native solutions using Azure, AWS, Databricks, and PySpark. Proven expertise in designing end-to-end data workflows processing high-volume datasets (100+ GB/day), with a strong focus on performance optimization and reliability. Designed and implemented enterprise-grade GenAI applications, including RAG-based chatbots using Azure OpenAI, integrating data engineering and delivering intelligent, context-aware solutions for enterprise use cases.

WORK EXPERIENCE

Data Engineer | ITC Infotech

Feb 2022 - Present

- Designed and implemented scalable data pipelines using **Azure Data Factory** and **Databricks**, processing large-scale datasets and improving data processing efficiency by **25%**
- Built scalable data transformations using **PySpark** and **Delta Lake**, improving performance by **35%**
- Developed event-driven workflows using **Azure Function Apps** and **Logic Apps**
- Implemented CI/CD pipelines using **Azure DevOps** for automated deployments
- Built and integrated **GenAI solutions** using LLM APIs for intelligent data querying, reducing manual effort by **30%**
- Implemented **RAG pipelines** using embeddings and vector search for document-based Q&A systems, reducing hallucinations in LLM responses
- Utilized **LangChain** for orchestrating LLM workflows and enabling context-aware multi-step reasoning
- Designed prompt engineering strategies to improve response accuracy and relevance
- Developed an interactive chatbot UI using **Streamlit** integrated with Azure OpenAI-based RAG pipeline
- Developed **agent-based systems** for task automation using tool integration and memory
- Managed and secured cloud infrastructure using **Azure Resource Groups, Key Vault, and Service Principals** for access control and secret management

Python Developer | Centrox Solution

Aug 2020 - Feb 2022

- Developed and deployed Python-based backend services using **Flask**
- Integrated APIs with **AWS RDS PostgreSQL** using SQLAlchemy
- Built scalable file upload systems using **Amazon S3** and **boto3**
- Deployed applications on **AWS EC2** using **Nginx** and **Gunicorn**
- Implemented secure authentication and payment systems using **Stripe** and **Plaid**
- Managed IAM roles and permissions for secure access control
- Containerized applications using **Docker** for portability and scalability
- Worked with **Linux (Ubuntu, CentOS)** environments for application deployment, server management, and troubleshooting

TECHNICAL SKILLS

- **Languages:** Python, SQL, PySpark
- **Cloud:** Azure (ADF, Databricks, Storage Accounts, Function Apps, Logic Apps, Key Vault, Resource Groups), AWS (EC2, S3, Lambda, IAM, CloudWatch)
- **GenAI:** LLM APIs, LangChain, Prompt Engineering, RAG, Agentic AI
- **Backend:** Flask, REST APIs
- **Databases:** SQL Server, MySQL
- **Tools:** Azure DevOps, Docker, Git, Linux, Postman, ServiceNow, PowerAutomate

ACHIEVEMENTS & CERTIFICATIONS

- Best Performer Award, ITC Infotech (2024)
- Preparing for Microsoft Certified: Azure Data Engineer Associate (DP-203)
- Preparing for Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Agile Management Fundamentals

EDUCATION

Master in Computer Applications

2017 - 2020

Silicon University, Bhubaneswar